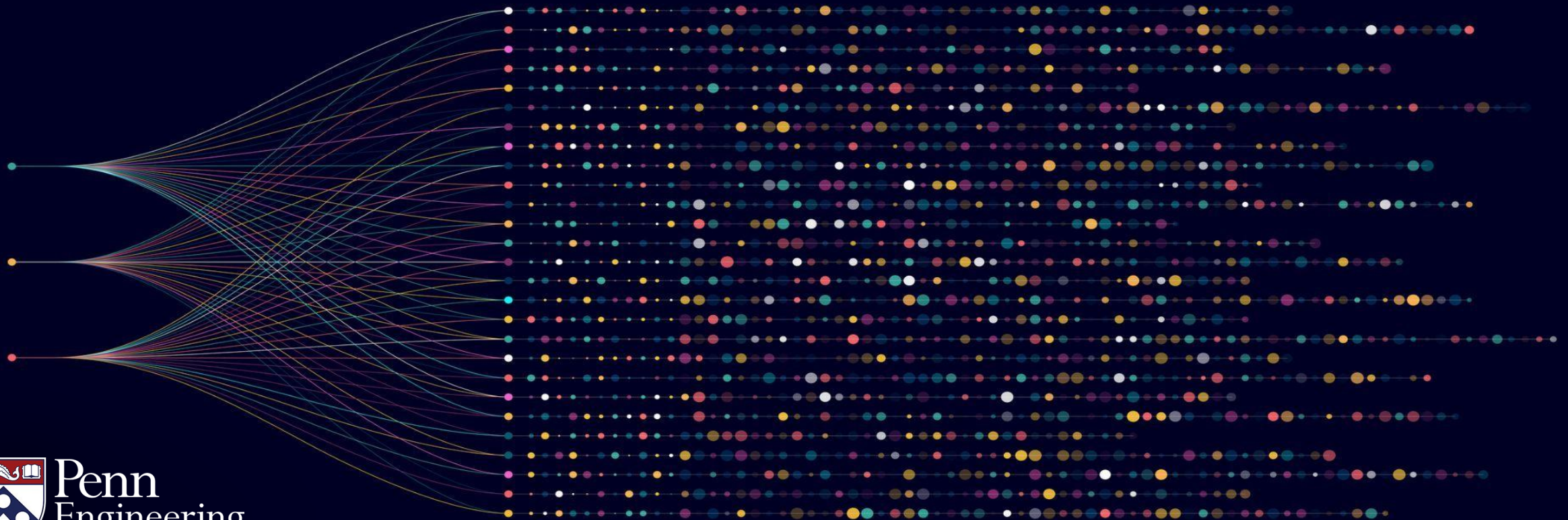


Algorithms for Big Data

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Additive Approximation Guarantees



Objectives: Computing Large Averages

- Estimating large averages
- Additive approximations

Defining the Computational Problem

Input: $X_1, \dots, X_n \in \mathbb{R}$

Goal: $S = \frac{1}{n} \sum_{i=1}^n X_i$

Queries: X_i

Examples of large averages:

1. How many users are in Europe?
2. How many users posted in the past week?
3. How long did users spend on the platform on average?

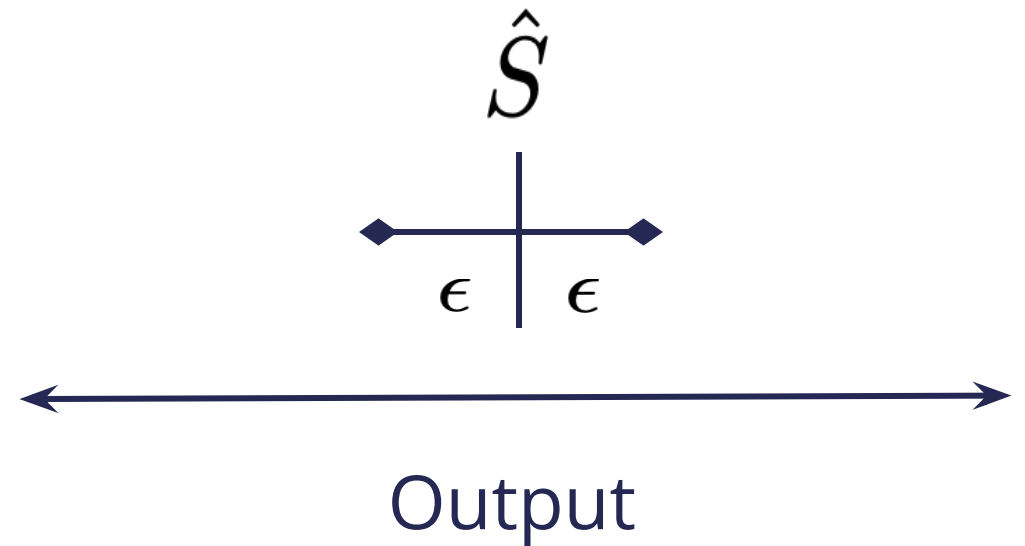
When is approximation adequate?

- Would an approximation to the output be sufficient?
- How would the result of the algorithm impact downstream decisions (or other algorithms)?
- If input is too large, no choice but to approximate. It is important to know what you are getting.

What do approximation guarantees look like?

- A number approximates another number when they are close
- Execute an algorithm which outputs $\hat{S} \in \mathbb{R}$ such that for a known $\epsilon \geq 0$:

$$|\hat{S} - S| \leq \epsilon$$



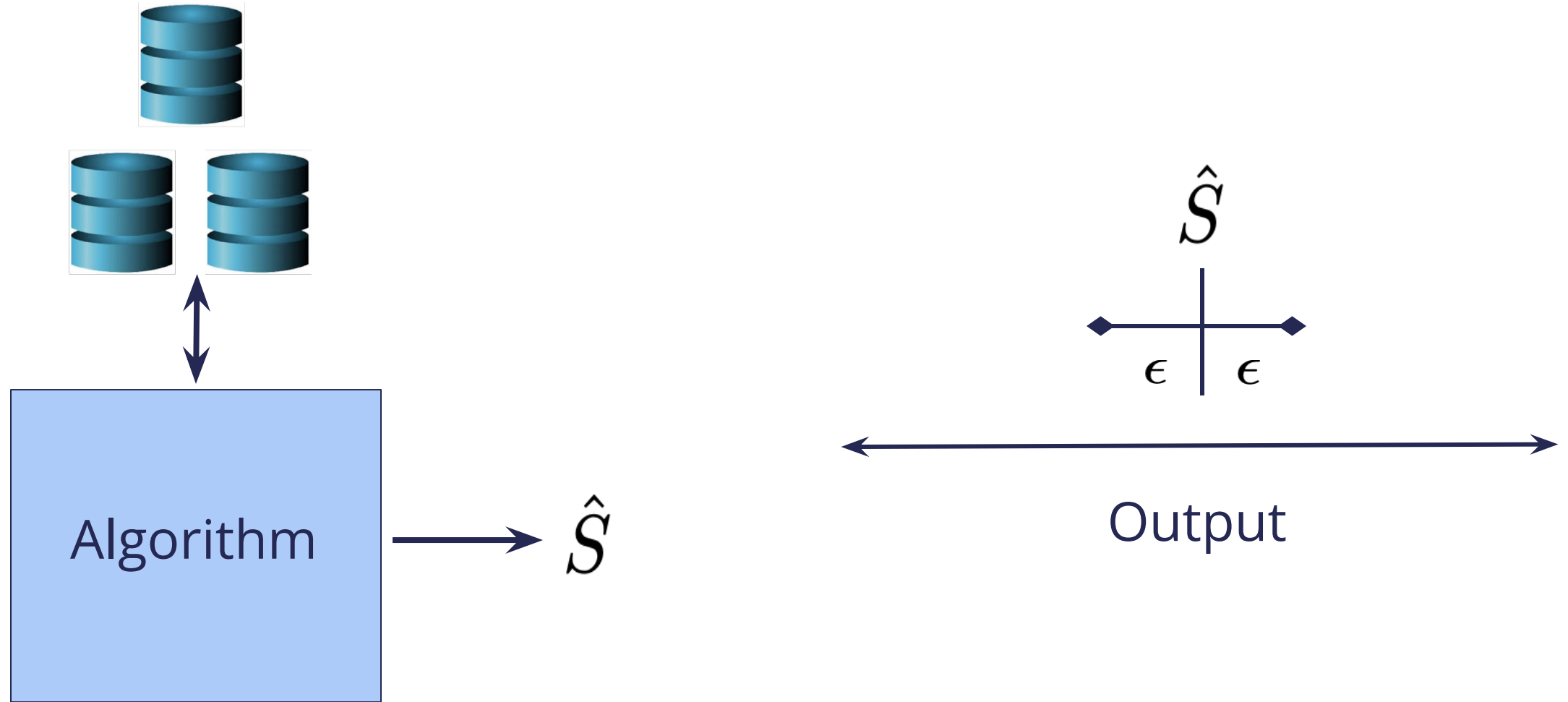
Additive ϵ -approximations

Definition: An algorithm for computing a large average is an additive ϵ -approximation if it satisfies the following guarantees:

- Given input query access to $X_1, \dots, X_n \in \mathbb{R}$.
- It produces an output number $\hat{S}$ which satisfies

$$\left| \hat{S} - \frac{1}{n} \sum_{i=1}^n X_i \right| \leq \epsilon.$$

What do approximation guarantees look like?



Interpreting results of approximations

- How many users are in Europe?
 - If accuracy is 0.05, and output is 0.3, then between 25% and 35% of users are in Europe
- How many users posted in the past week?
 - If accuracy is 0.1 and output is 0.9, then between 80% and 100% of users posted in the past week
- How long (on average) did users spend on the platform today?
 - If accuracy is 30 and output is 120, then between 90 and 150 seconds

Approximation guarantees

- Accuracy parameter $\epsilon \geq 0$. As ϵ becomes smaller, the algorithm becomes more accurate. When zero, algorithm is exact.
- As the algorithm becomes more accurate, the number of queries grows. More accurate algorithms are more expensive.
- Big sacrifice. Algorithm will no longer be exactly correct.

Key algorithmic questions:

- Do there exist (more) efficient algorithms when allowing approximation?
 - By more efficient, fewer than $O(n)$ queries, which was the brute force benchmark.
- If so, what is the tradeoff between the number of queries and the accuracy parameter.
- Turns out no, unless algorithm is also randomized.